

Emergency-Priority Protocol for V2X and High-Loss Networks

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99.0%

Emergency delivery
at 30% WAN loss

+40%

More DENMs delivered
vs DSRC 802.11p

88×

Lower p99 latency
vs MQTT

2.8 ms

Emergency p99
across all densities

Urban Intersection Results (N = 75 agents, 5-run Monte Carlo, seed = 42)

Channel: Two-ray LOS + WINNER+ B1 NLOS · CBR ≈ 0.36 · DENM deadline: 50 ms

Protocol	DENM PDR	DENM < 50 ms	BSM PDR	Emerg. p99
[*] QDAP	75.5%	75.5%	66.2%	2.8 ms
802.11bd	73.4%	73.4%	67.6%	1.5 ms
C-V2X Mode 4	67.2%	33.2% [MISS]	61.1%	100.5 ms
DSRC 802.11p	53.8%	53.8%	56.3%	2.5 ms
UDP	65.8%	65.8%	59.0%	8.9 ms
MQTT	59.3%	48.2% [MISS]	53.3%	247.1 ms

1. The Problem

HTTP/1.1 & HTTP/2

Head-of-line blocking — retransmit lost packets, stall new ones.
At 30% WAN loss: only 68.8% of messages meet 500 ms deadline.

MQTT

Broker-mediated with equal priority. Emergency alerts queue behind telemetry. TCP RTO ~200 ms. p99 reaches 247 ms in V2X.

DSRC / 802.11p

No application-layer priority. Collision-warning DENM waits behind routine BSMs. 15% collision probability at CBR = 0.50.

C-V2X Mode 4

Semi-Persistent Scheduling adds 0–100 ms per transmission regardless of urgency. 67% of DENMs miss the ETSI 50 ms safety deadline.

2. QDAP Architecture

QFT Scheduler

Emergency messages jump the queue via log-linear softmax priority. 374,000 scheduling decisions/sec. Zero head-of-line blocking.

Adaptive FEC

Observes real-time channel loss per source. Emergency traffic: up to 4× coded redundancy. $P(\text{fail}) = \text{per}^k$, fire-and-forget.

Ghost Session

0-RTT reconnection after link drop. No TCP handshake overhead. Critical for vehicles moving at 130 km/h between RSUs.

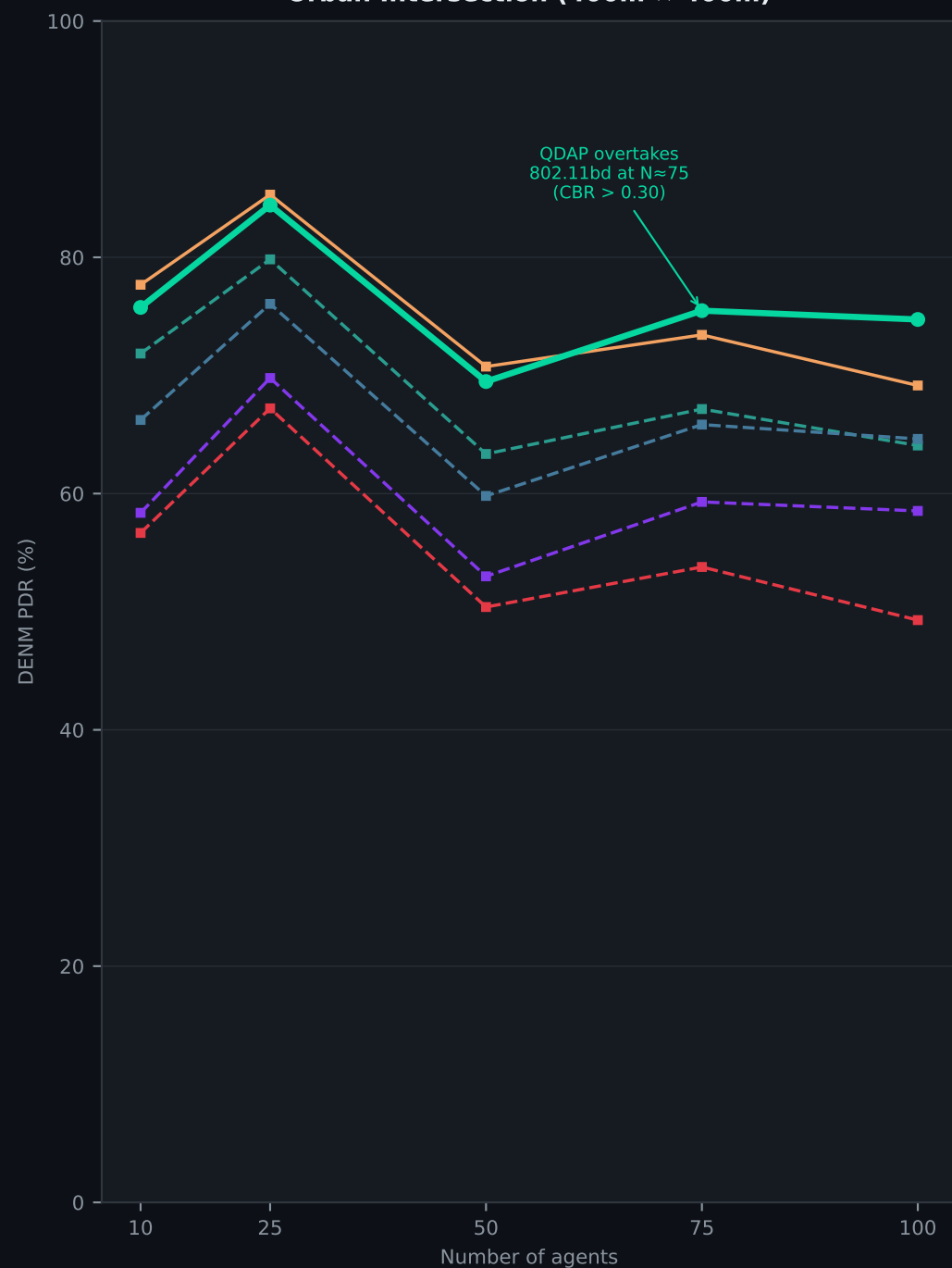
Delta Encoder

Position updates transmitted as tiny deltas vs last known state. 74.4% BSM size reduction. Effective PHY throughput: 20 Mbps.

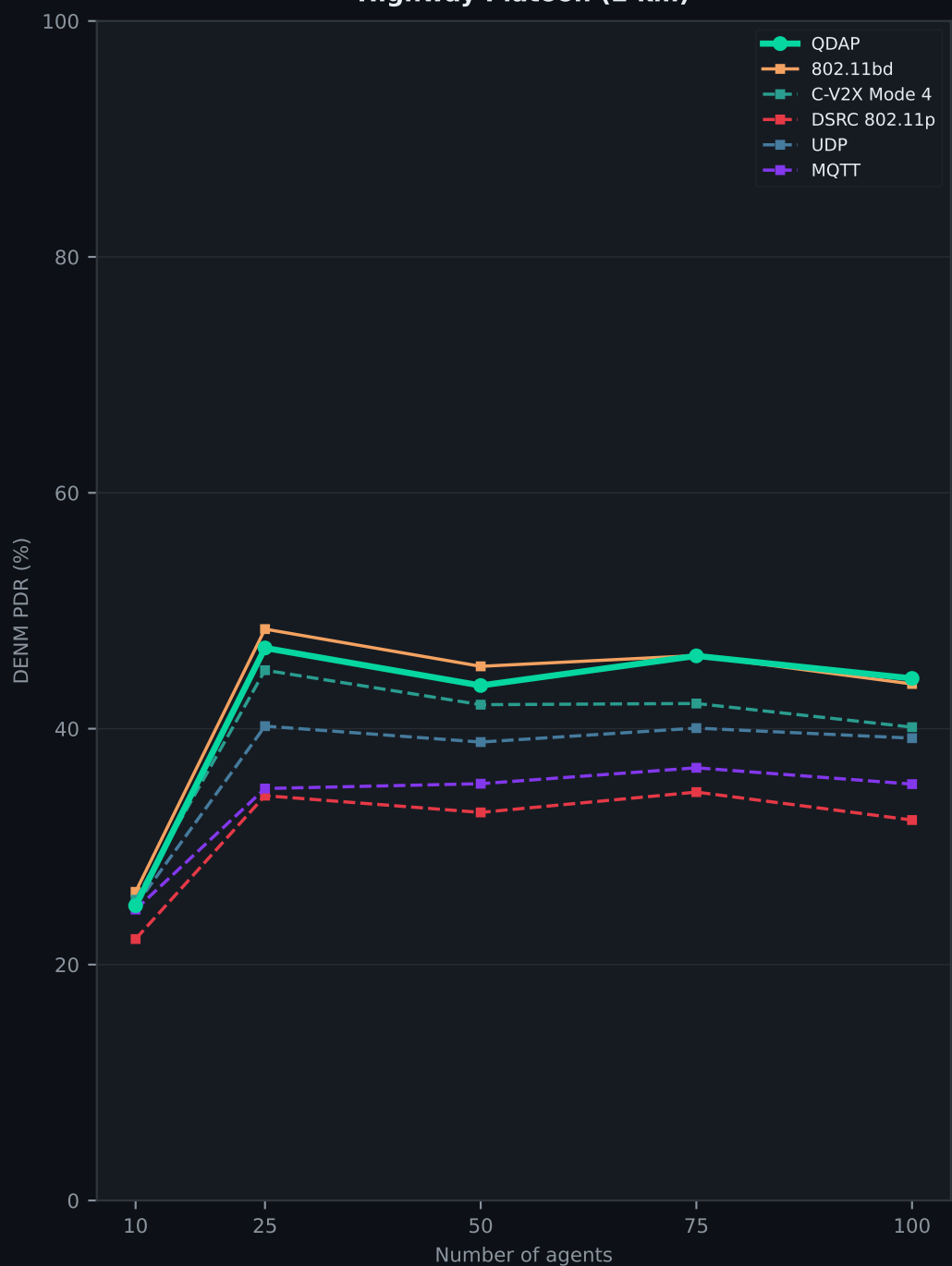
Protocol Stack	
QFT Scheduler	→ Emergency first, 374k decisions/s
Adaptive FEC	→ Up to 4× redundancy, fire-and-forget
Ghost Session	→ 0-RTT reconnect, moving vehicles
Delta Encoder	→ 74.4% BSM compression
TCP / UDP	→ No hardware changes required

DENM PDR vs Vehicle Density — 5-run Monte Carlo, seed = 42

Urban Intersection (400m × 400m)

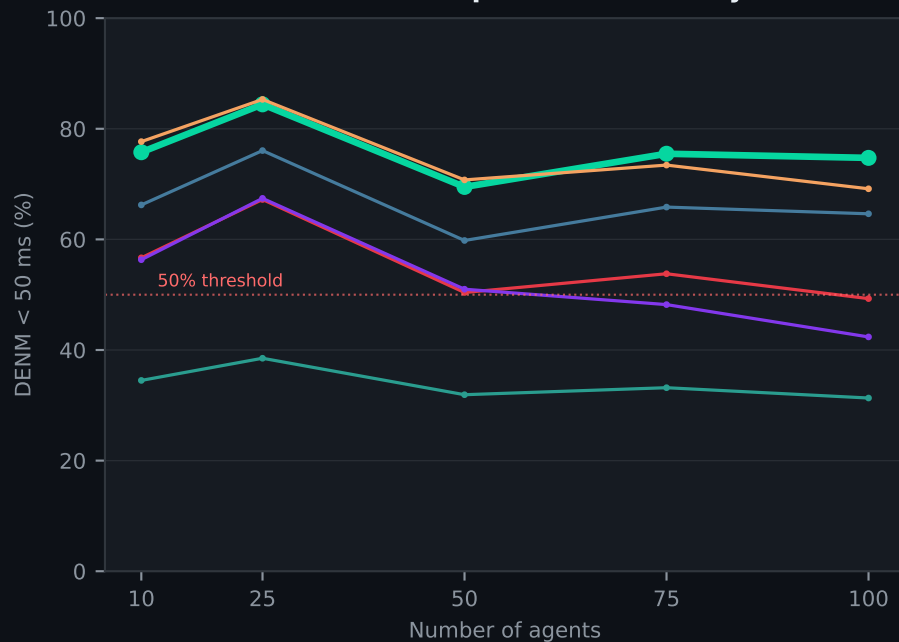


Highway Platoon (2 km)

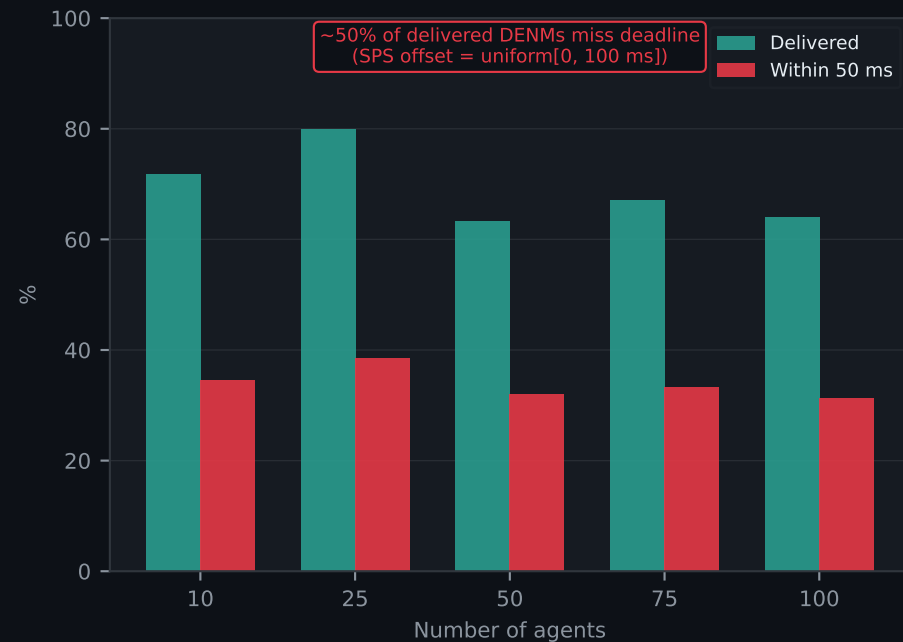


DENM Within 50 ms Safety Deadline — Urban Intersection

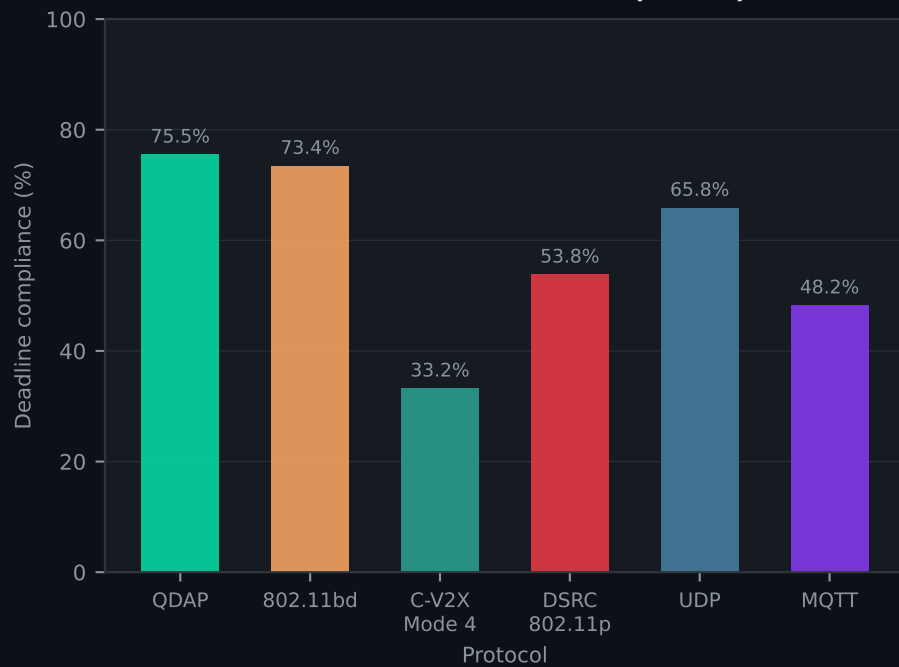
Deadline Compliance vs Density



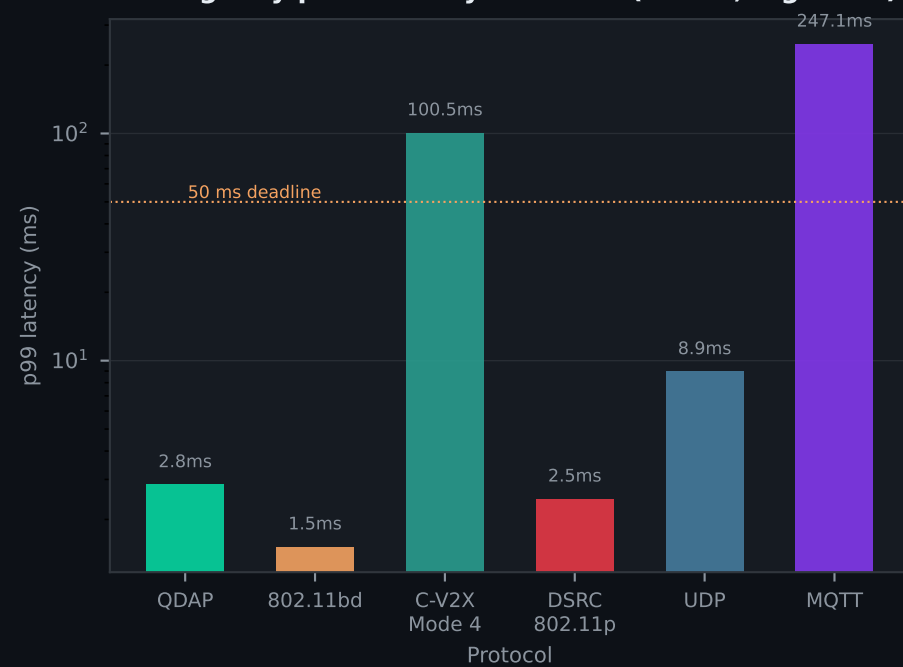
C-V2X Mode 4 — SPS Deadline Leak



DENM < 50 ms at N=75 (Urban)



Emergency p99 Latency at N=75 (Urban, log scale)

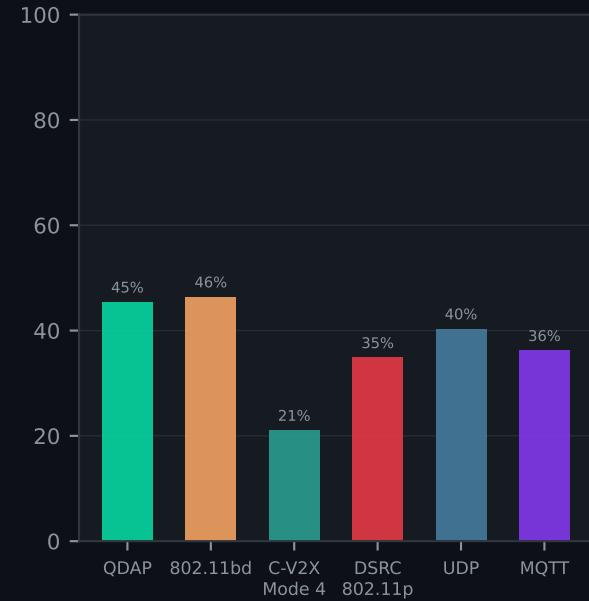
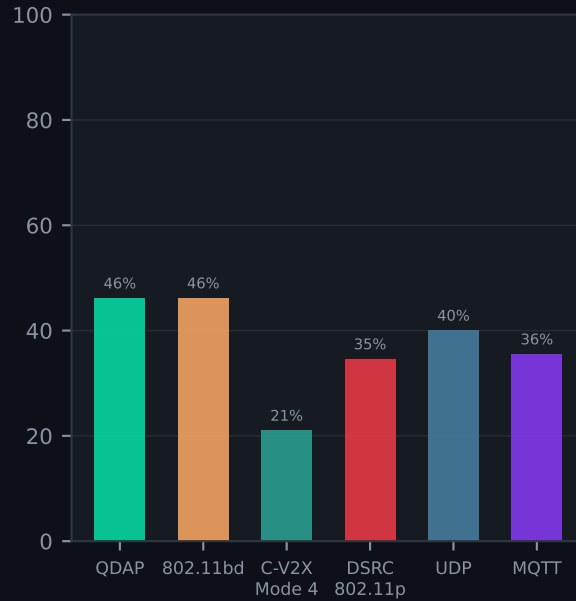
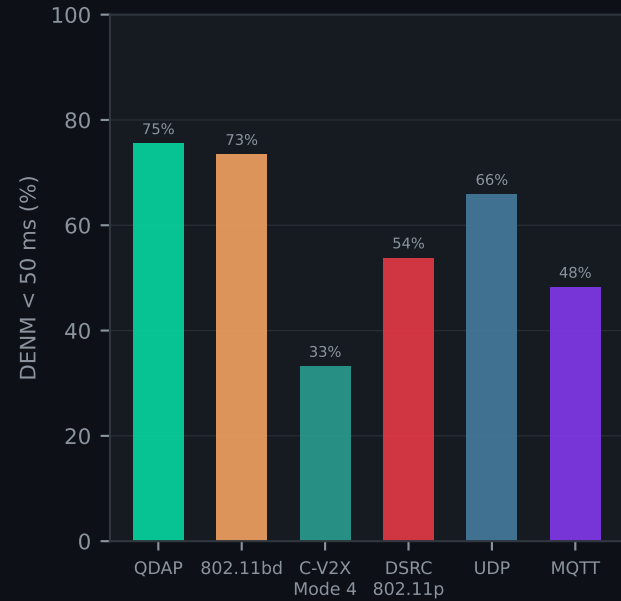
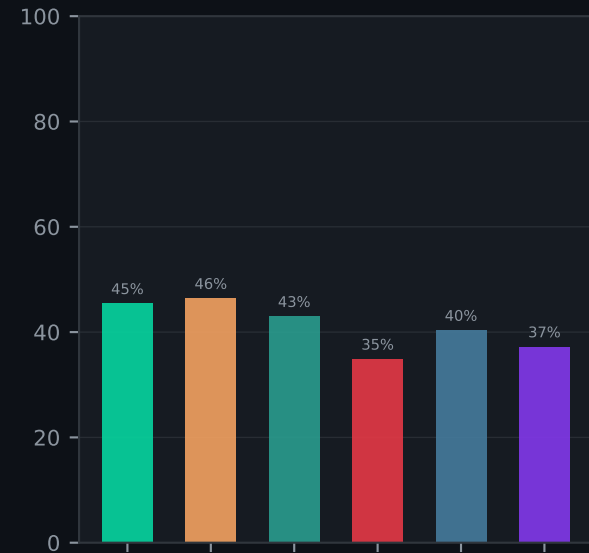
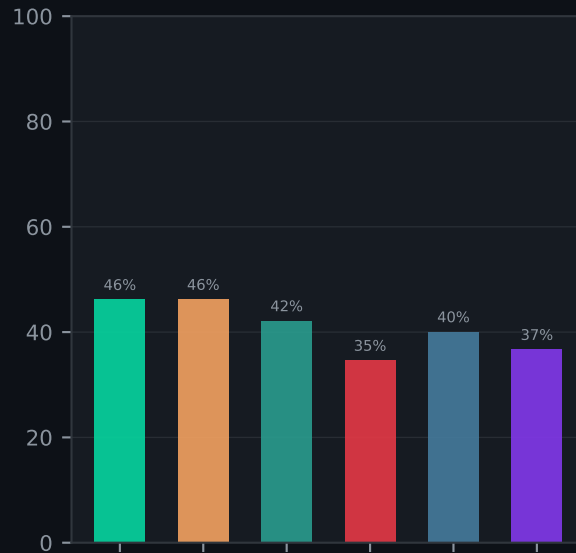
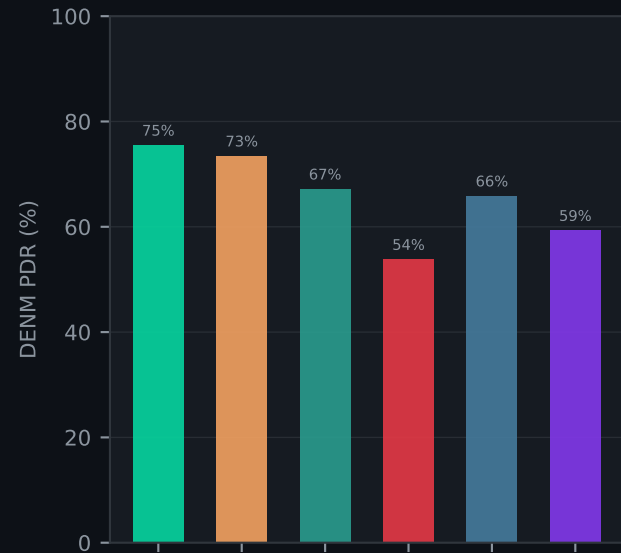


Protocol Comparison Across All Scenarios — N = 75 agents

Urban Intersection

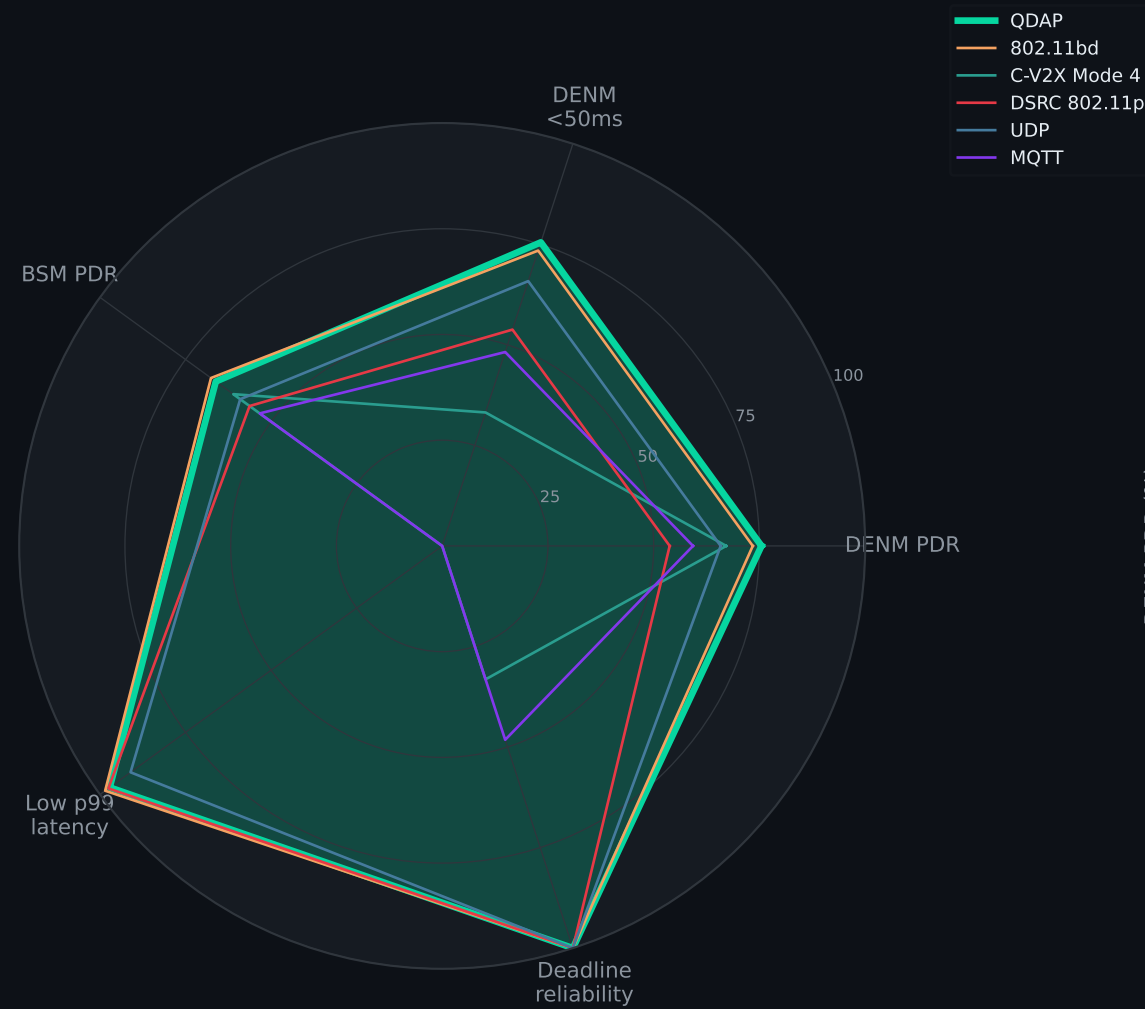
Highway Platoon

Emergency Cascade



QDAP 802.11bd C-V2X Mode 4 DSRC 802.11p UDP MQTT

Multi-Metric Protocol Radar — Urban N=75



Highway Platoon — DENM PDR vs Density



Simulation Methodology & Limitations

Channel Model

- LOS: Two-ray ground reflection ($h_t = h_r = 1.5\text{ m}$)
 $PL = 40 \cdot \log_{10}(d) - 7.04\text{ dB}$ (validated: Gozalvez 2012)
- NLOS: WINNER+ B1 $PL = 36.7 \cdot \log_{10}(d) + 22.7 + 26 \cdot \log_{10}(f_c)$
- Shadowing: log-normal $\sigma = 4\text{ dB}$ LOS, 8 dB NLOS
- Fading: Nakagami-m ($m=2$ LOS, $m=1$ Rayleigh NLOS)
- SNR→PER: sigmoid fit to published BER curves per PHY

Traffic & Scenarios

- Urban: $400\text{m} \times 400\text{m}$, 4 corner buildings ($170\text{m} \times 170\text{m}$ AABB)
70% cars · 20% motorcycles · 10% pedestrians (VRU priority)
- Highway: 2 km dual-lane, 100–130 km/h, 85% cars / 15% motos
- Cascade: Highway + pedestrian hazard at $x=500\text{m}$
- CBR ≈ 0.50 urban @ $N=100$ · ≈ 0.53 highway @ $N=100$

Message Types

- BSM (SAE J2735): 400 B · 10 Hz · 100 ms deadline
- DENM (ETSI EN 302 637): 250 B · event-driven · 50 ms deadline
- Emergency injection: deterministic at step 25% and 65%
(same across all protocols — fair comparison guaranteed)

Monte Carlo Setup

- 5 independent runs per (scenario, density, protocol)
- Seeds: $\text{base_seed} + \text{run_idx} \times 1000$ (reproducible)
- Densities: $N \in \{10, 25, 50, 75, 100\}$ agents
- 450 total combinations (5 densities × 6 protocols × 3 scenarios × 5 runs)
- 20 s simulated time per run · 50 ms timestep

Protocol Models

- DSRC: ICPR-based collision $p_{\text{col}} = \text{CBR}^2 \times 0.6$ (empirical broadcast)
- 802.11bd: +3 dB LDPC gain, better EDCA tuning
- C-V2X: SPS offset = uniform[0, 100 ms] (3GPP TS 36.213)
- MQTT: $P(\text{tcp_fail}) = 1 - (1 - \text{PER})^2$ QoS 1 two-hop
- QDAP: $P(\text{fail}) = \text{PER}^k$ with $k = 1.5\text{--}4.0$ adaptive

Limitations & Future Work

- V2X results are simulation-based — not real vehicle hardware
- QDAP currently uses TCP; ETSI ITS-G5 PHY integration = future work
- Multi-hop DENM relay not yet implemented
- LOS: simplified distance threshold (350 m); full ray-cast = future work
- Real-world validation on OBU/RSU hardware = next milestone